

SURAJ KIRAN AIRI

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EDUCATION

Chandigarh University

Bachelor of Engineering in Computer Science and Engineering, CGPA: 8.65

Mohali, India

2026

TECHNICAL SKILLS

GenAI & Retrieval: RAG, Hybrid Search (Vector + BM25), Reranking, RRF, Metadata Filtering, RAGAS Evaluation

LLM Frameworks: LangChain, LangGraph, LiteLLM, MCP, LangSmith, Transformers, OpenAI, Anthropic

Backend: Python, FastAPI, Node.js, Express.js, REST APIs

Databases: PostgreSQL, MongoDB, FAISS, ChromaDB

Infrastructure & Tools: Docker, Linux, Git, Postman, Jupyter

Specialization Certificates: IBM Generative AI Engineering (Coursera)

PROJECTS

Hope: AI Agent Harness Platform | *Python, FastAPI, Redis, PostgreSQL, S3, Node.js, Next.js* June 2026

- Developed a reusable AI agent harness platform that abstracts production concerns—including streaming, execution monitoring, usage tracking, and error alerting—allowing developers to focus exclusively on agent logic.
- Architected a three-layer execution engine (SDK / Runner / Engine) with strict boundary separation, enabling LangGraph, MCP/RAG connectors, LangSmith tracing, and future providers to integrate without modifying core execution logic.
- Engineered a stateful LLM execution lifecycle supporting Redis checkpointing, interrupt/resume, HITL workflows, per-step token/cost accounting, and fire-and-forget, synchronous, and SSE streaming execution modes.

Zensho: AI Video Editor | *Python, FastAPI, LiteLLM, Deepgram, React, OpenTimelineIO, Electron* March 2026

- Architected a **dual-environment system** with a Python/FastAPI native backend and Electron frontend, using IPC bridge for low-latency communication between inference layer and UI.
- Developed an LLM-powered inference pipeline using Deepgram and LiteLLM to generate word-level editing decisions for silences, filler words, stutters, and poor takes.
- Designed **privacy-first BYOK architecture** (all media processed locally) with **OpenTimelineIO export** for direct handoff to DaVinci Resolve and Premiere Pro, keeping AI decisions fully auditable and user-correctable.

EduPath: Production RAG Chatbot | *Python, FastAPI, LlamaCloud, Vector DB, BM25, RAGAS* May 2026

- Built a production-grade RAG pipeline using LlamaCloud parsing, metadata-aware indexing, and filtered retrieval to improve document-grounded response quality across education datasets.
- Implemented hybrid retrieval combining dense vector search, BM25, Reciprocal Rank Fusion (RRF), and cross-encoder reranking, improving document retrieval relevance.
- Evaluated retrieval quality using RAGAS (0.87 Context Precision, 0.89 Recall, 0.93 Faithfulness, 0.90 Answer Relevancy) and integrated LangSmith for end-to-end tracing and observability.

EXPERIENCE

Freelance Gen AI

March 2025 – Present

- Automated end-to-end lead generation using n8n and Vapi for US based client, replacing manual prospecting with scraping, qualification, and AI-driven outbound voice workflows.
- Deployed a RAG-powered assistant that automated routine course and enrollment queries, reducing manual support workload for institute staff.

Flutter Developer Intern

July 2023 – Sept. 2024

YoursThatSenior & SN Silos (2 paid internships)

Remote & Hybrid

- Delivered and deployed cross-platform Flutter applications across two production products, owning feature implementation, API integration, payment workflows, testing, and release.

ACHIEVEMENTS

Hackathons: Winner of **Smart India Hackathon (SIH) 2024**, Software Edition.

Competitions: Ranked **443/3800+** (Top 12%) in Kaggle ML Competition; Won Work Force 2023, IEEE CIS CUSB.

Open Source: Published **3 Python libraries on PyPI** – ai-tools-executor & visualkit.

Technical: Attended and taught **10+ ML sessions** at university; Multiple ML certifications.